

HealthConnect Clinic

Data Preparation, Feature Engineering & Baseline Model Development

AnalystLab Africa Experience Lab - Week 5 Track: Data Science

SUBMISSION REPORT

1. Introduction

Week 5 focused on moving the HealthConnect project from planning into practical machine learning development.

The main objective was to prepare the appointment dataset and build an initial model for predicting patient appointment no-shows.

The purpose of this stage was not to create the final model. The goal was to establish a reliable baseline that can be improved in later stages.

2. Week 4 Problem Review

During Week 4, the HealthConnect business problem was translated into a supervised machine learning problem.

The proposed task was binary classification.

The model aims to predict whether a scheduled appointment will result in:

- Attended
- No-Show

Cancelled appointments were treated separately because cancellation represents a different behaviour from failing to attend.

Potential predictors were identified from patient information, appointment details, booking behaviour, previous appointment history, reminder information, distance to the clinic, and estimated waiting time.

Week 4 also identified important concerns such as data leakage, repeated patients, missing values, prediction timing, and model fairness.

Week 5 built directly on these decisions.

3. Data Preparation

The dataset was reviewed before modelling.

The main preparation steps included checking:

- Missing values
- Duplicate records
- Data types
- Target definition
- Irrelevant variables
- Potential leakage

3.1 Missing Values

Three variables contained missing values.

`reminder_channel` had missing values when no reminder had been sent.

These values were replaced with "No Reminder".

This was preferred over statistical imputation because the missing values had a clear meaning.

Missing values in `distance_to_clinic_km` and `waiting_time_minutes` were replaced using their median values.

Median imputation was selected because it is less sensitive to extreme observations than the mean.

3.2 Duplicate Records

The dataset was checked for duplicate rows.

No duplicate records were identified.

3.3 Data Types

Variable types were reviewed before analysis.

Date variables were converted to datetime format where required.

Categorical and numerical variables were also identified for later preprocessing.

4. Target Preparation

The original `appointment_outcome` variable contained three outcomes:

- Attended
- No-Show
- Cancelled

Cancelled appointments were removed from the modelling dataset.

The final prediction problem therefore focused on:

Attended vs No-Show

The target variable was encoded using `LabelEncoder`.

The final coding was:

- 0 = Attended
- 1 = No-Show

After cancellations were removed, the two target classes were approximately balanced.

This meant that major class-imbalance correction was not required for the baseline model.

5. Exploratory Data Analysis

EDA was performed to understand the variables and identify patterns that could influence modelling decisions.

The analysis included:

- Target distribution
- Numerical distributions

- Numerical variables against appointment outcome
- Categorical variables against appointment outcome
- Outlier assessment
- Correlation analysis

5.1 Target Distribution

Attended and No-Show appointments were approximately balanced.

This was useful because the model could be trained without a severe imbalance between the two outcome classes.

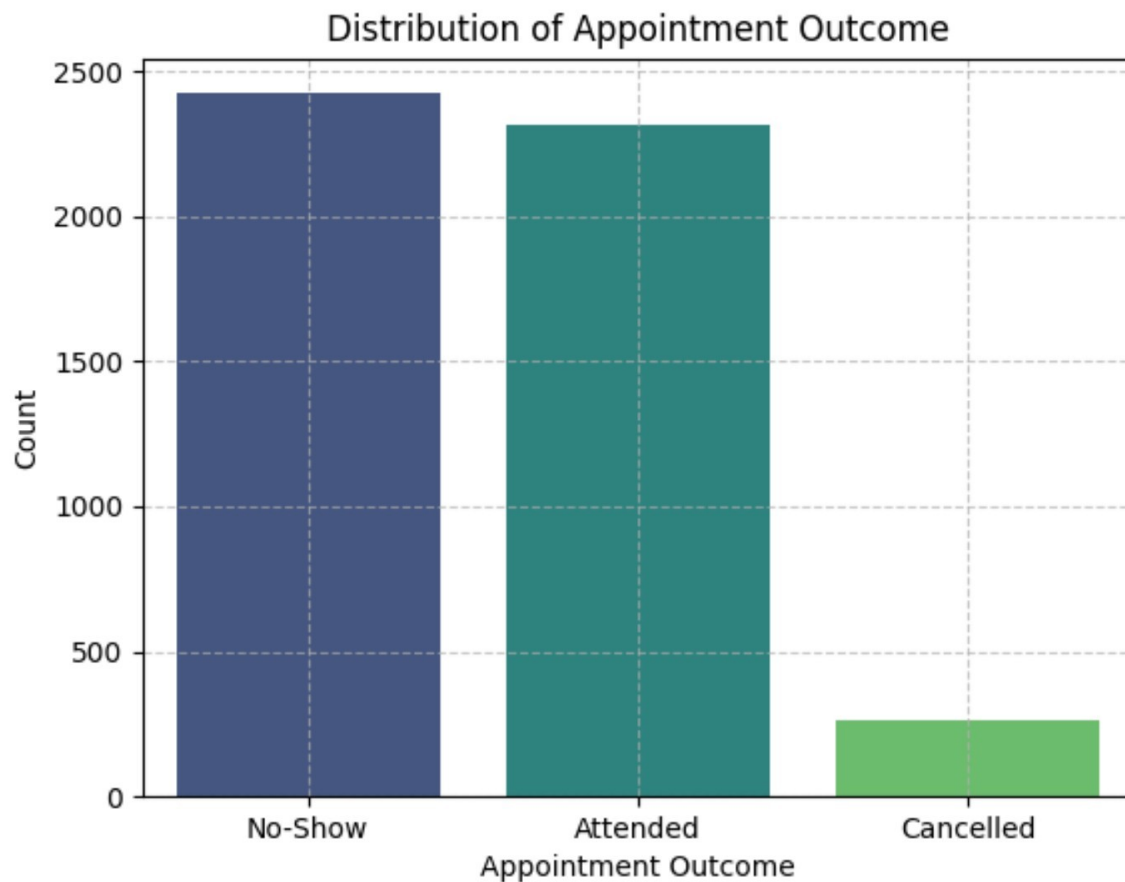


Figure 2. Distribution of Appointment Outcome

5.2 Numerical Variables

Booking lead time showed one of the clearest differences between the two appointment outcomes. No-show appointments generally had longer booking lead times than attended appointments.

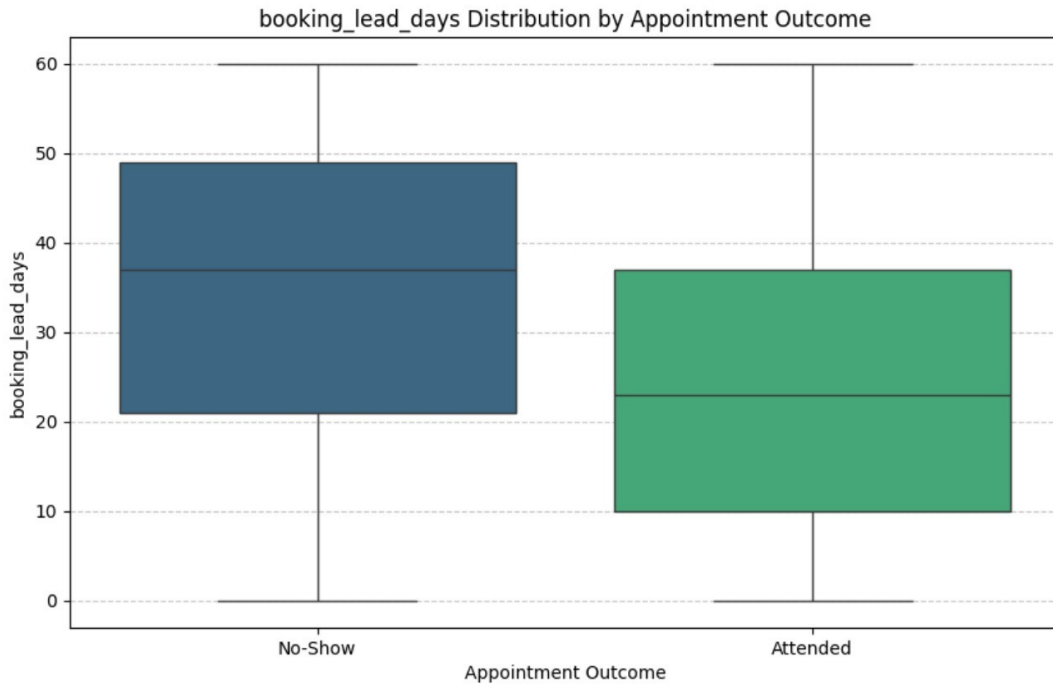


Figure 2. booking_lead_days Distribution by Appointment Outcome

Age showed substantial overlap between the two groups.

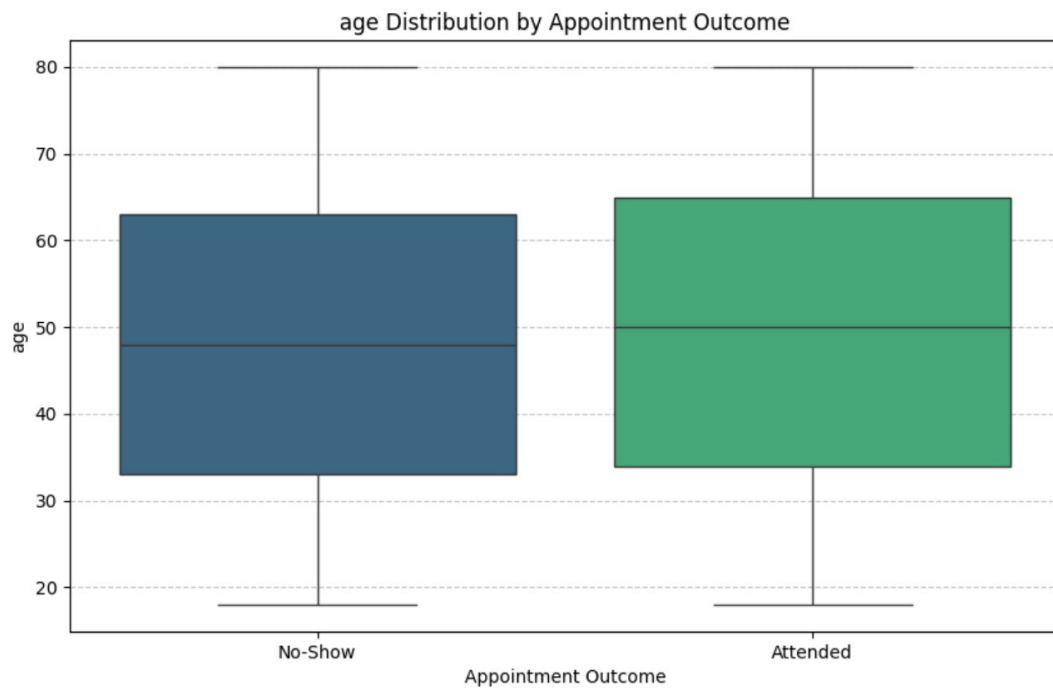


Figure 3. age Distribution by Appointment Outcome

Previous appointments and previous no-shows also showed considerable overlap when examined individually.

Distance to the clinic was right-skewed and contained several high values.

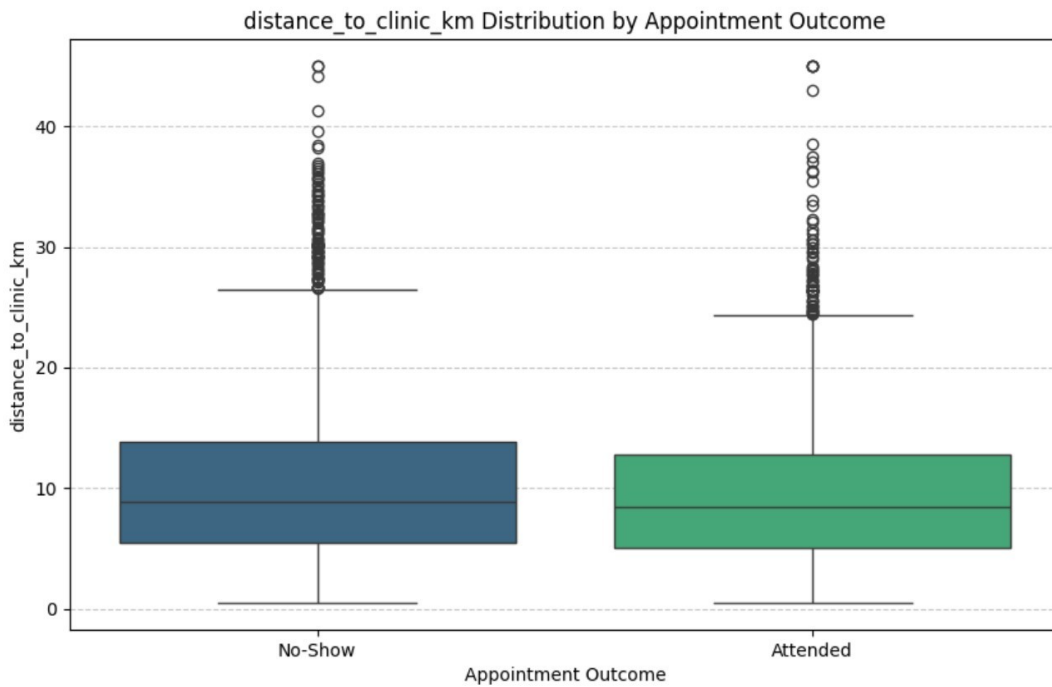


Figure 4. distance_to_clinic_km Distribution by Appointment Outcome

Waiting time had a more central distribution with some higher observations.

These extreme observations were not automatically removed because they were not clearly identified as data errors.

5.3 Categorical Variables

Categorical variables showed different levels of association with appointment outcome.

Appointment day showed some variation in attendance patterns.

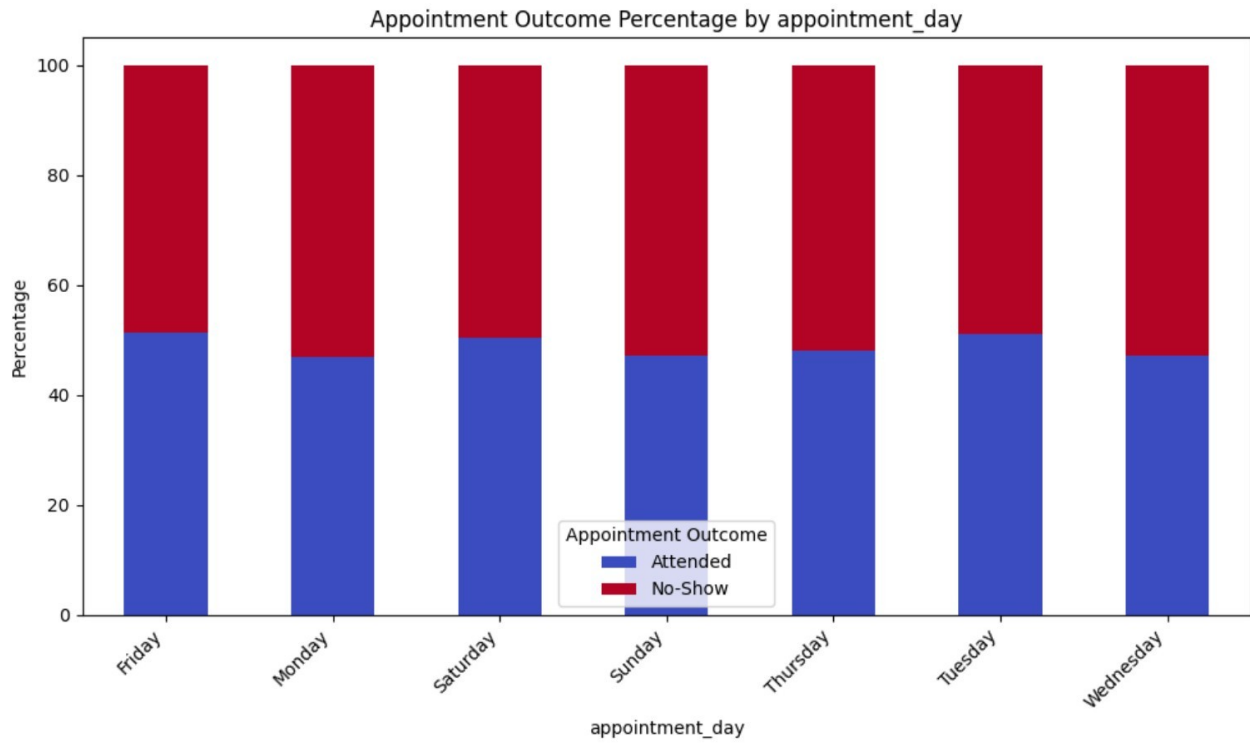


Figure 5. Appointment Outcome Percentage by appointment_day

Reminder information also showed differences between categories.

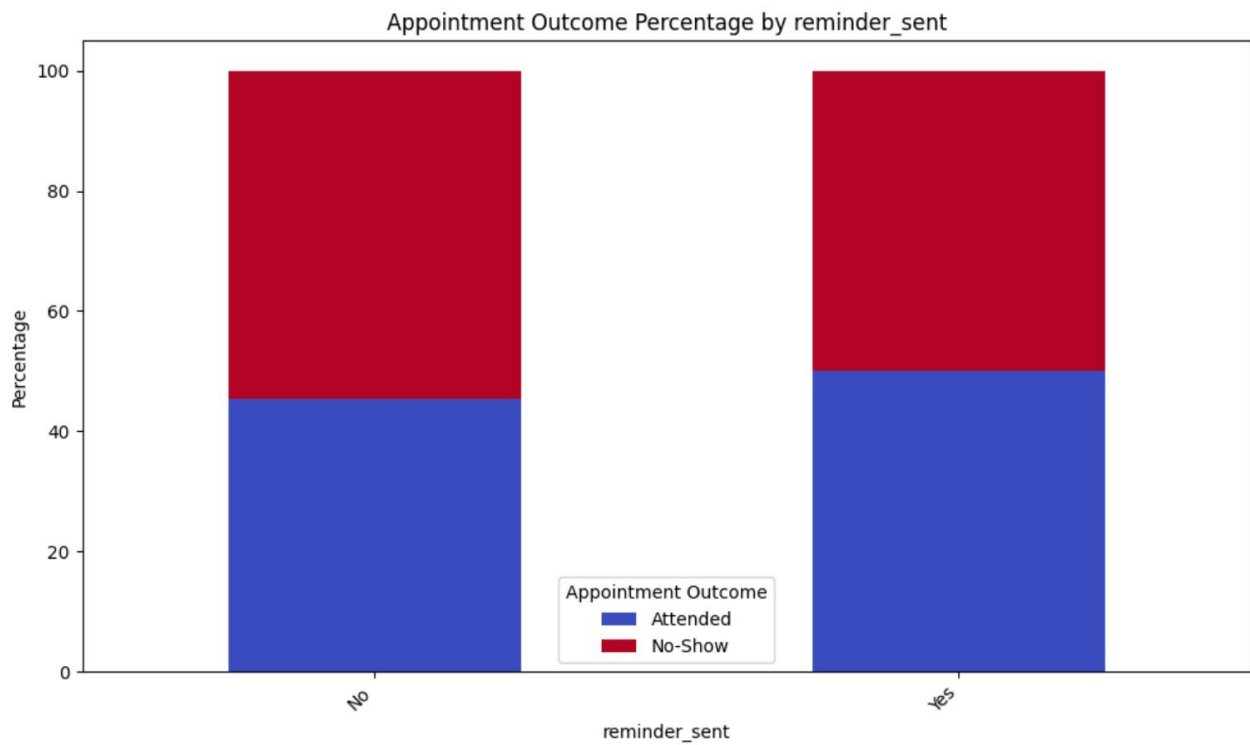


Figure 6. Appointment Outcome Percentage by reminder_sent

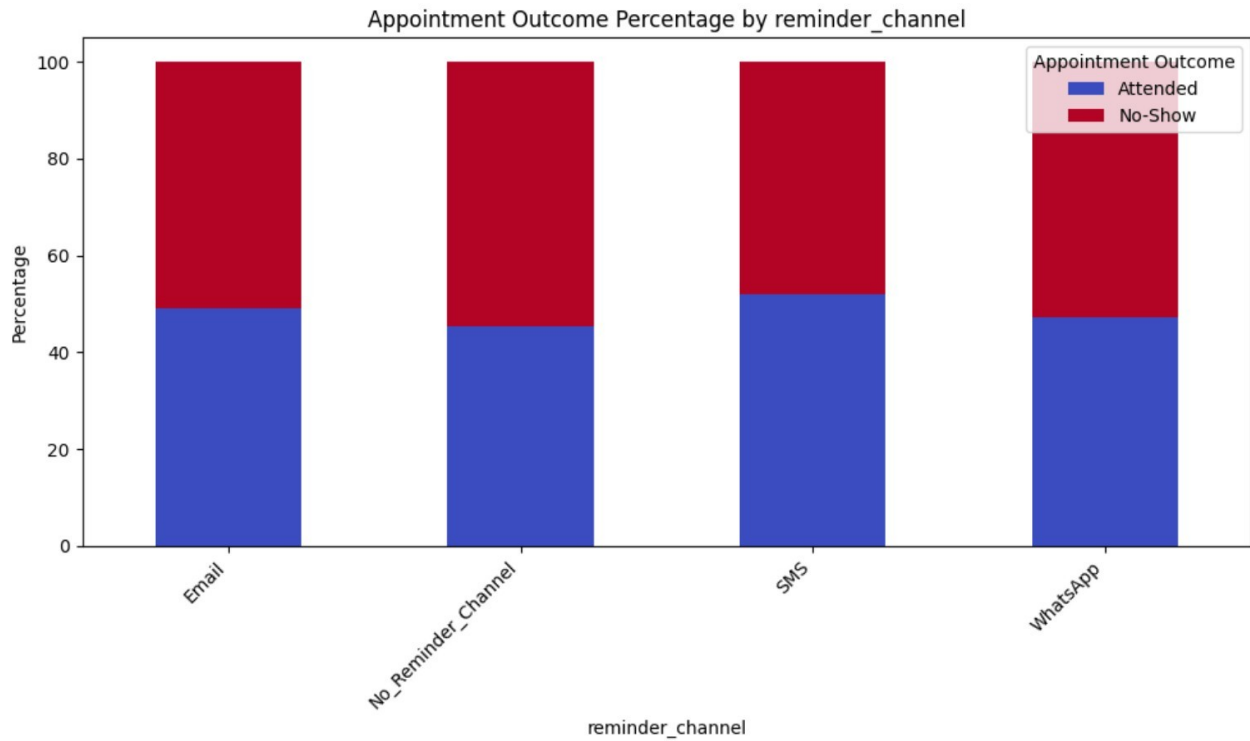


Figure 7. Appointment Outcome Percentage by reminder_channel

However, many categorical variables showed only modest differences when considered individually.

They were still retained as candidate predictors because a weak individual relationship does not mean a variable cannot contribute within a multivariable model.

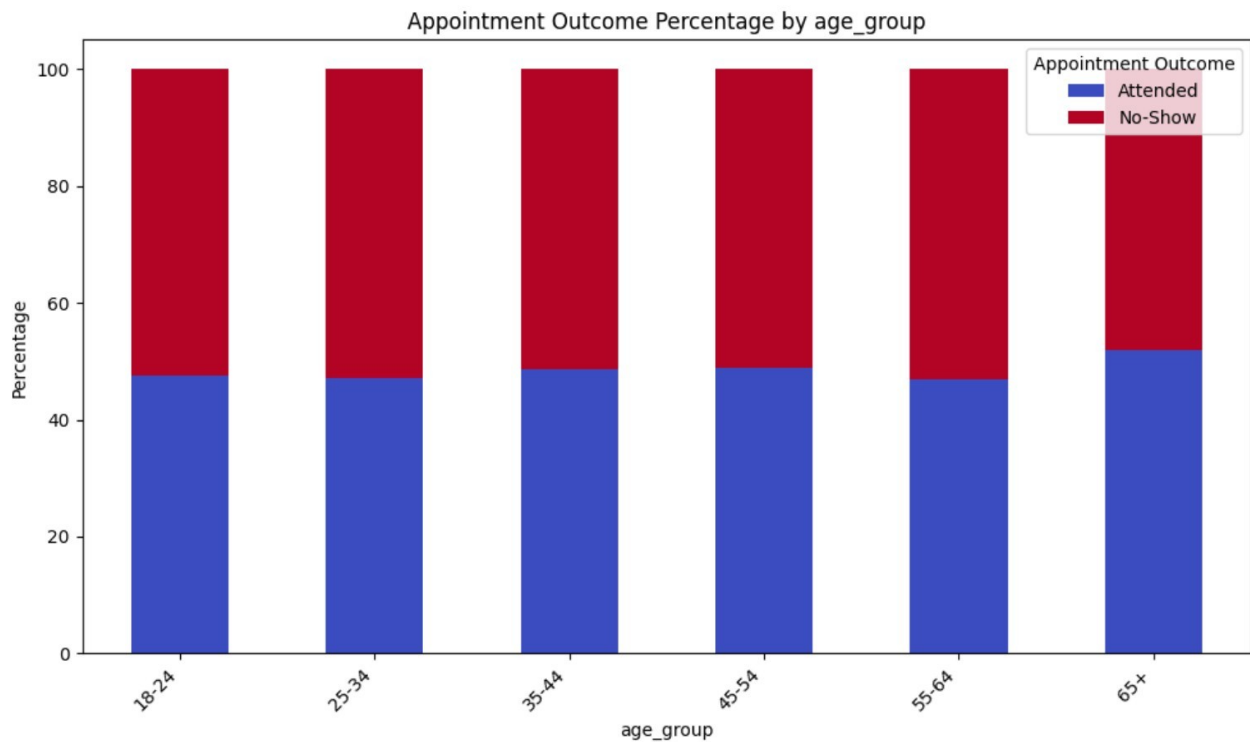


Figure 8. Appointment Outcome Percentage by age_group

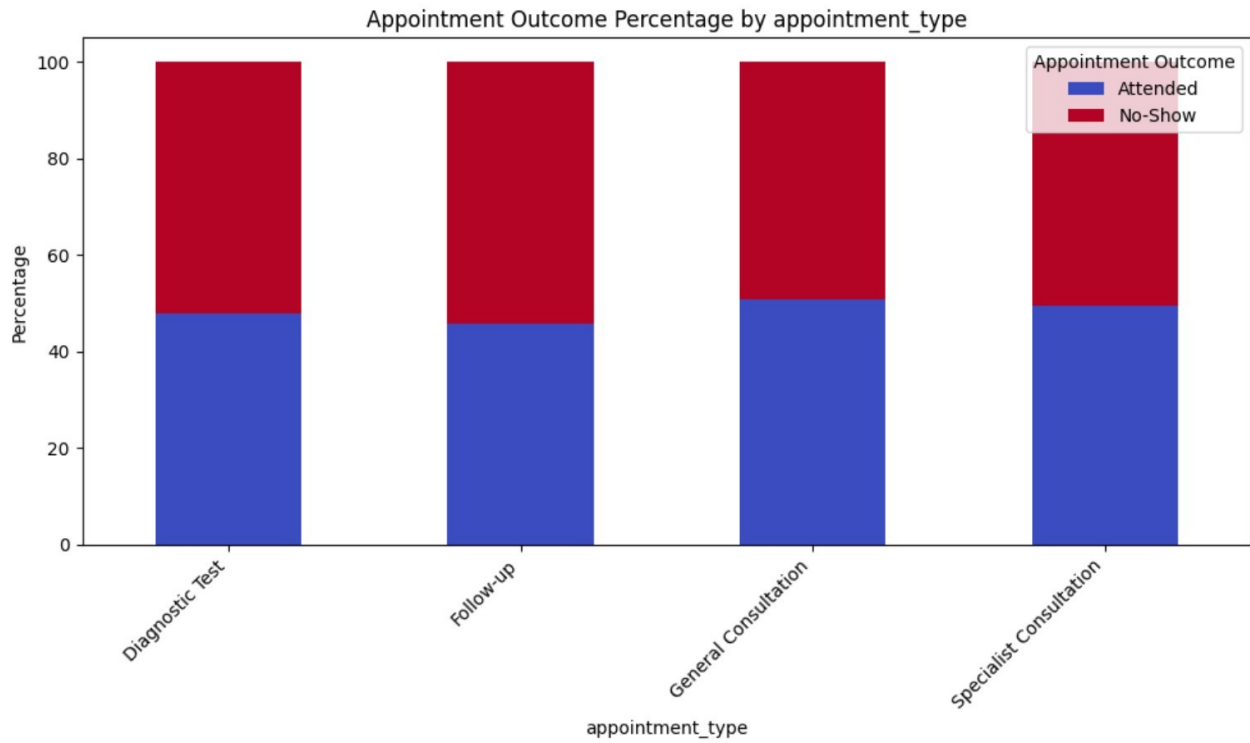


Figure 9. Appointment Outcome Percentage by appointment_type

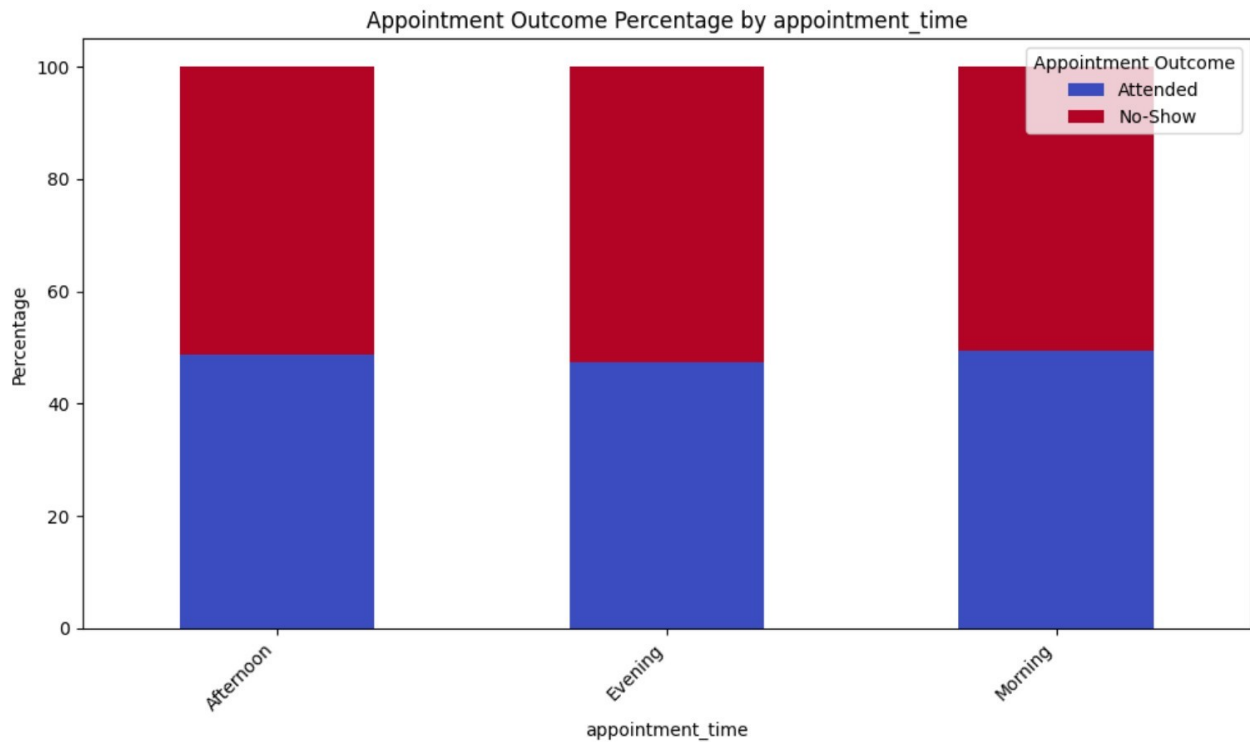


Figure 10. Appointment Outcome Percentage by appointment_time

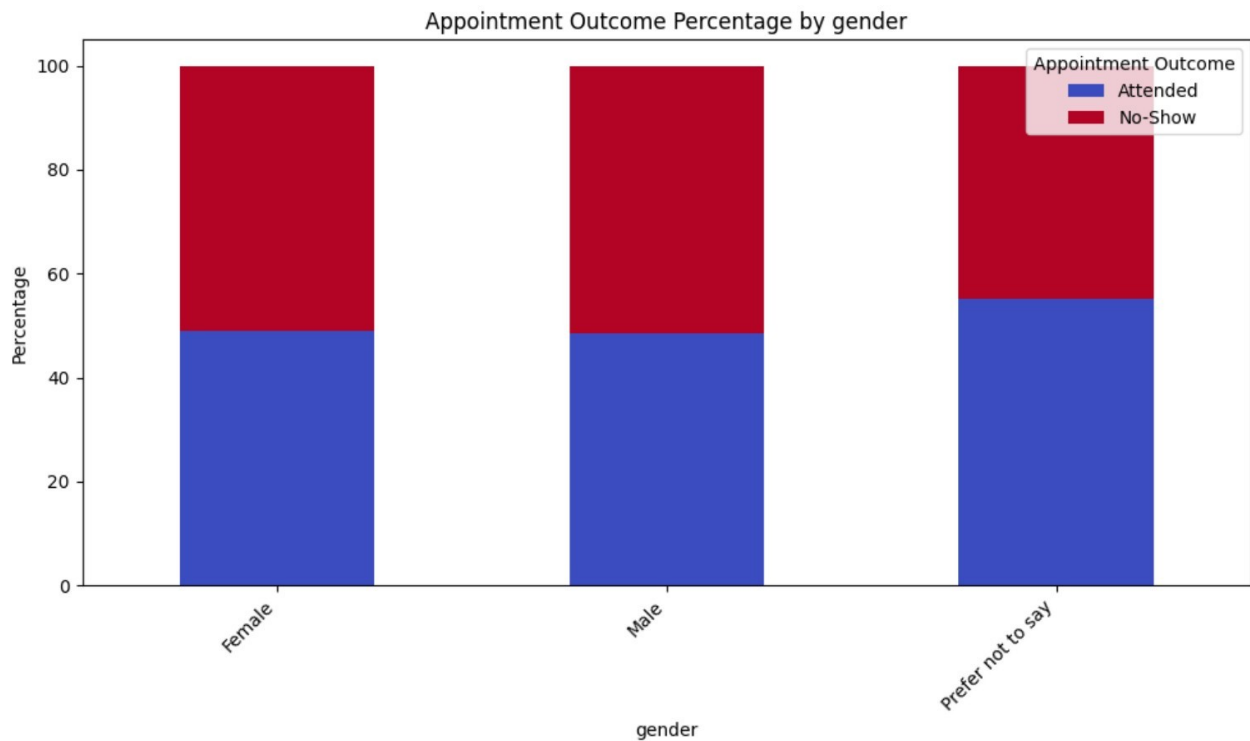


Figure 11. Appointment Outcome Percentage by gender

6. Correlation Analysis

Correlation analysis was performed on the numerical features.

Most numerical predictors showed weak correlations with each other.

A moderate positive correlation was observed between `previous_appointments` and `previous_no_shows`.

No severe multicollinearity was initially identified among the original numerical predictors.

Correlation was examined again after feature engineering to identify redundant engineered variables.

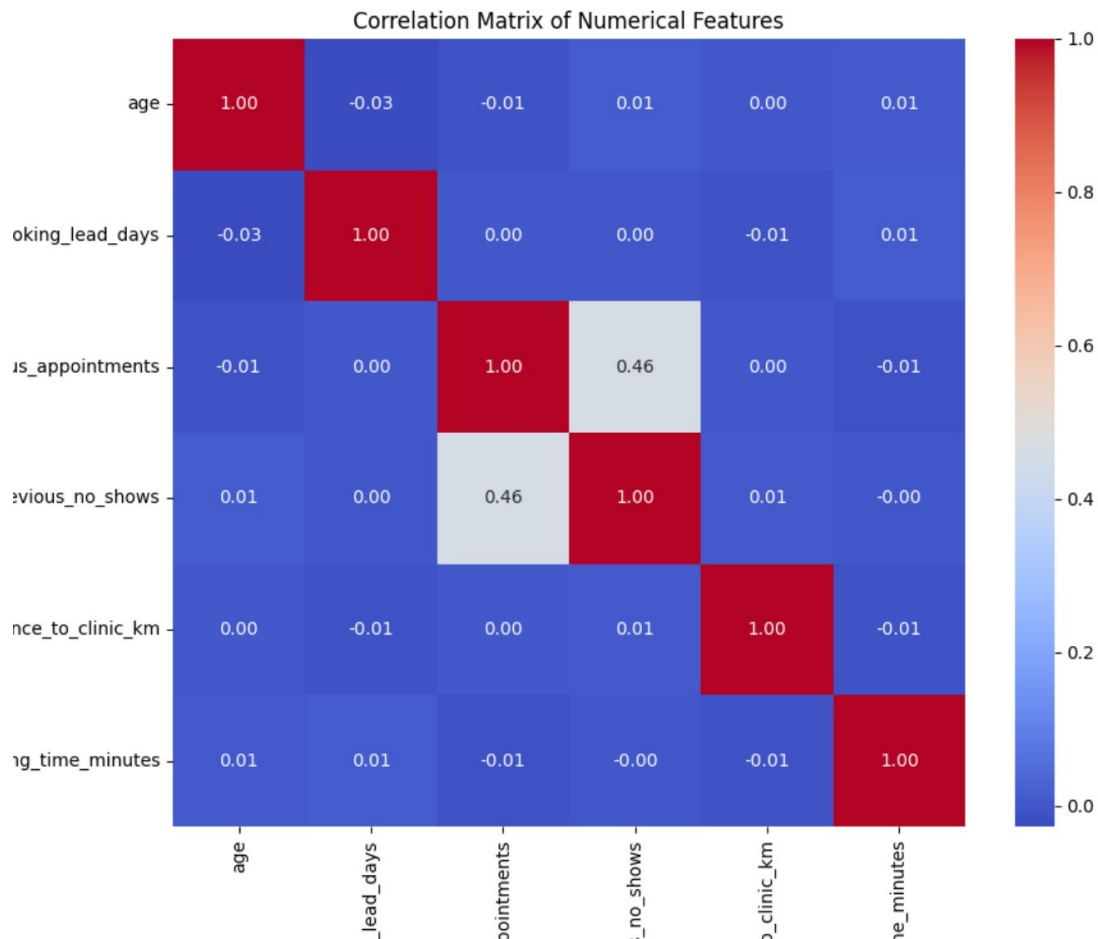


Figure 12. Correlation Matrix of Numerical Features

7. Feature Engineering

Several features were created to better represent patient behaviour and appointment characteristics.

7.1 Previous No-Show Rate

A `previous_no_show_rate` feature was created.

It represented the proportion of previous appointments that resulted in a no-show.

This provides more context than the raw number of previous no-shows.

For example, one previous no-show out of one appointment represents a different history from one no-show out of ten appointments.

7.2 Previous No-Show Indicator

A `has_previous_no_show` feature was created.

It identified whether a patient had ever previously missed an appointment.

7.3 Long Booking Lead

A `long_booking_lead` indicator was explored to represent appointments with longer booking lead times.

However, this variable was strongly correlated with the original continuous `booking_lead_days` variable.

The continuous variable preserved more information and was therefore preferred for baseline modelling.

7.4 Booking Month

A `booking_month` feature was extracted from the booking date.

This was created to allow the model to capture possible seasonal patterns in appointment booking behaviour.

8. Feature Assessment

The engineered features were assessed for redundancy.

Several strong correlations were identified.

`long_booking_lead` had a correlation of approximately 0.86 with `booking_lead_days`.

`has_previous_no_show` was also strongly related to previous no-show variables.

`previous_no_show_rate` provided a more informative representation of previous attendance behaviour and was retained.

Redundant variables were excluded where appropriate to reduce unnecessary multicollinearity in the Logistic Regression baseline.

This process helped keep the baseline model relatively simple and interpretable.

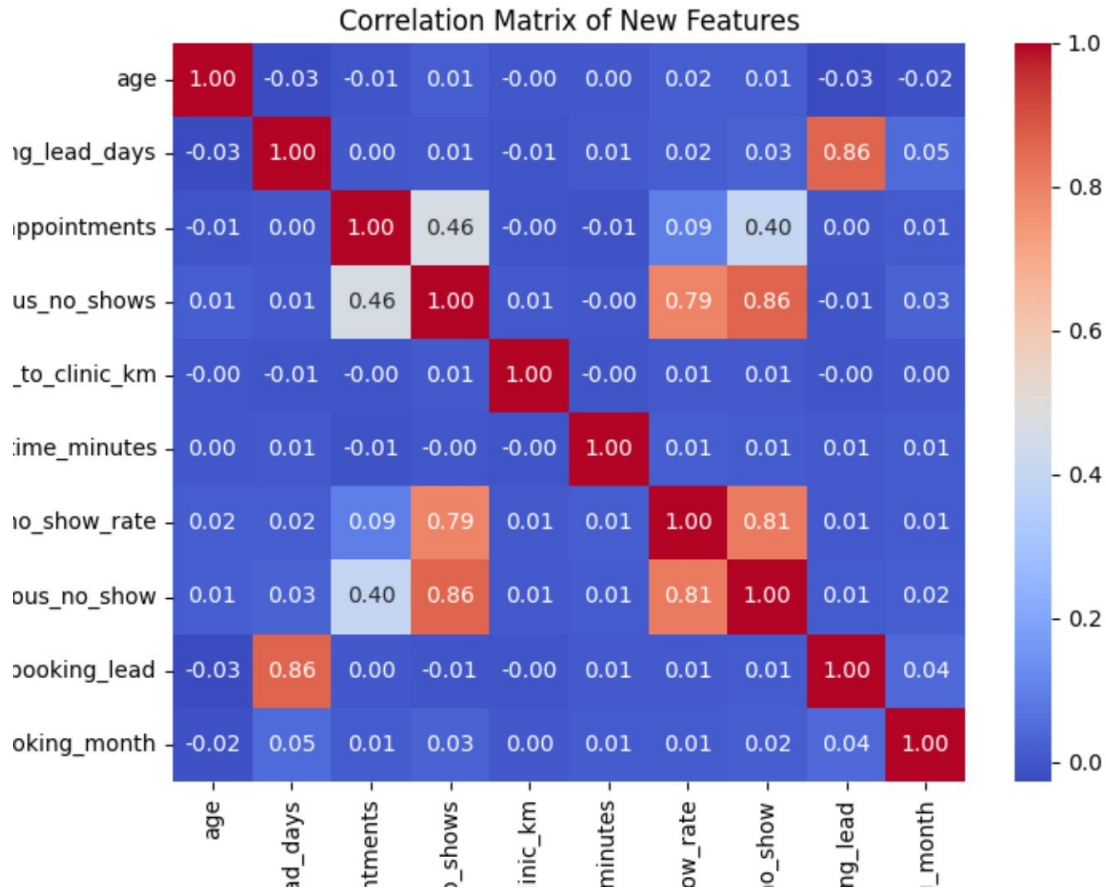


Figure 13. Correlation Matrix of New Features

9. Train/Test Strategy

A normal random row-level split was not considered appropriate.

The dataset contains repeated appointments belonging to the same patients.

If appointments from the same patient appeared in both the training and testing datasets, the evaluation could become less realistic.

A GroupShuffleSplit was therefore used.

patient_id served as the grouping variable.

An approximately 80/20 train-test split was created.

The resulting datasets contained approximately:

- 3,773 training appointments
- 966 testing appointments

A validation check confirmed that there were:

- 0 patients shared between the training and testing sets.

This provided a more realistic evaluation of how the model might perform on unseen patients.

10. Data Preprocessing

The predictor variables contained both numerical and categorical information.

Categorical predictors were encoded into a numerical form suitable for Logistic Regression.

Numerical variables were scaled where appropriate.

Preprocessing was applied with the training and testing structure in mind to reduce the risk of information leakage.

The target variable remained separately encoded as a binary outcome.

11. Baseline Model

Logistic Regression was selected as the baseline classification algorithm.

It was chosen because it is simple, widely used for binary classification, and relatively interpretable.

The purpose of this model was to establish an initial performance benchmark.

More complex algorithms can later be compared against this baseline.

12. Model Evaluation

The baseline Logistic Regression model was evaluated using several metrics.

Metric	Result
Accuracy	0.63
No-Show Precision	0.62
No-Show Recall	0.65
No-Show F1-Score	0.64
ROC-AUC	0.68

Accuracy alone was not used to judge the model.

Recall was particularly important because the project aims to identify appointments that may become no-shows.

13. Confusion Matrix

The confusion matrix showed:

Actual Outcome	Predicted Attended	Predicted No-Show
Attended	292	191
No-Show	168	315

The model correctly identified 315 actual no-show appointments.

It missed 168 actual no-shows.

It also incorrectly classified 191 attended appointments as no-shows.

These results show that the baseline model contains useful predictive information but still makes a substantial number of errors.

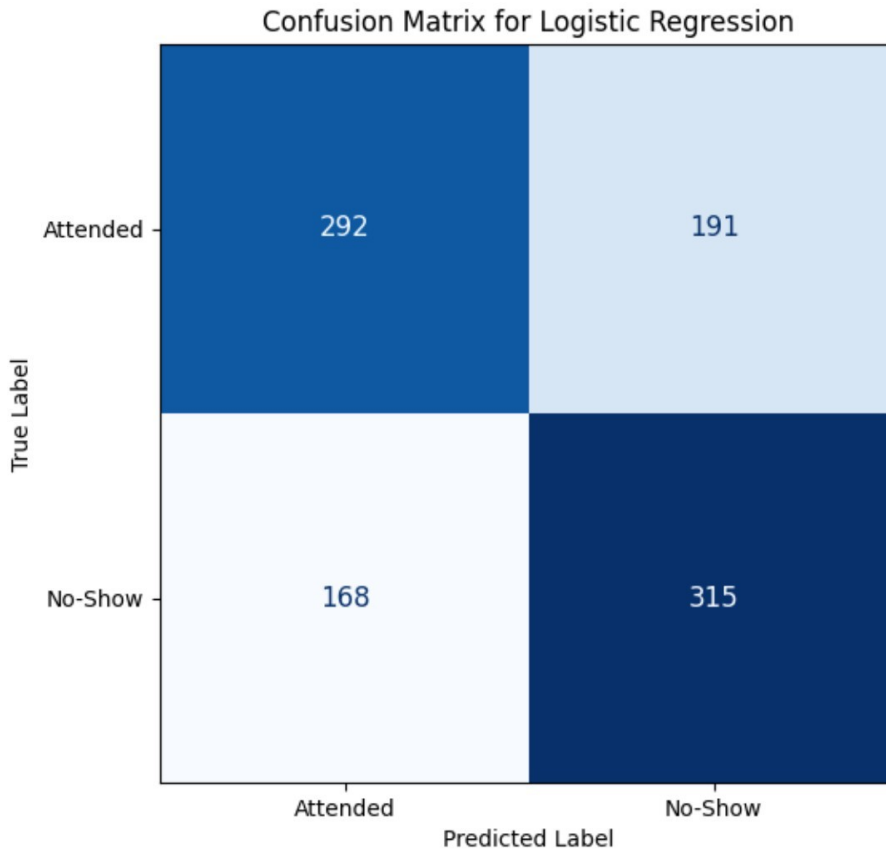


Figure 14. Confusion Matrix for Logistic Regression

14. ROC-AUC

The Logistic Regression model achieved a ROC-AUC of 0.68.

The ROC curve remained above the random-classification reference line.

This indicates that the model has moderate ability to distinguish between attended and no-show appointments.

The result is suitable as an initial baseline.

Further improvements should be explored in later stages.

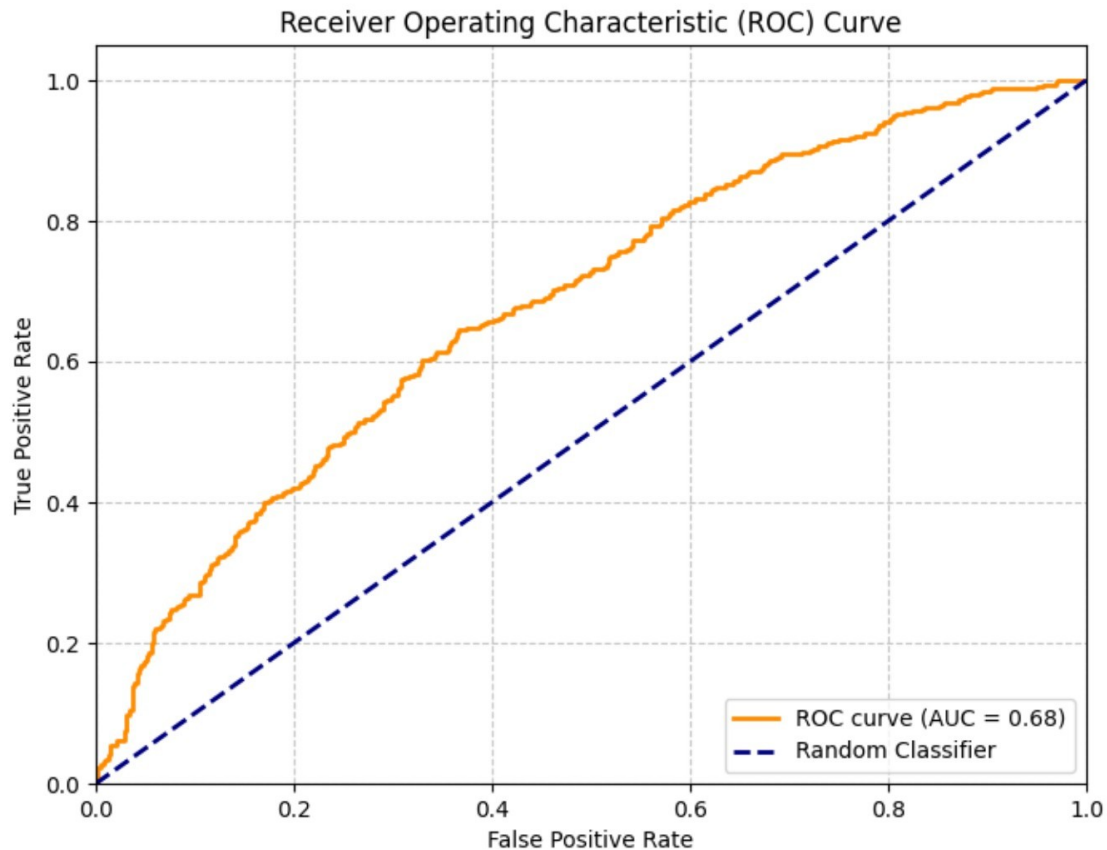


Figure 15. Receiver Operating Characteristic (ROC) Curve

15. Interpretation of the Baseline Model

The baseline model achieved an overall accuracy of approximately 63%.

More importantly, it identified approximately 65% of actual no-show appointments.

The No-Show F1-score was 0.64, showing a reasonable balance between precision and recall.

The ROC-AUC of 0.68 also indicates that the available appointment data contains useful predictive signal.

However, the model is not yet strong enough to be considered a final solution.

Its main value is to establish a benchmark against which future models can be compared.

16. Cross-Track Collaboration

Cross-track collaboration was completed with a Data Analytics intern.

The purpose of the interaction was to compare operational findings from Data Analytics with the variables identified during Data Science analysis.

The Data Analytics findings highlighted:

- Reminder status
- Previous no-show history
- Booking lead time

These factors showed associations with appointment attendance and no-show rates.

The findings supported several observations from the Data Science analysis.

Booking lead time had already shown one of the clearest differences between attended and no-show appointments.

Previous no-show behaviour was also represented through the engineered `previous_no_show_rate` feature.

The collaboration provided an independent operational perspective and strengthened the interpretation of the selected predictors.



Alia Al-Qadri Yesterday at 8:01 PM

Hi OSOWO

I'm working on the Data Science track for HealthConnect and have completed the initial EDA and baseline no-show prediction model. From my analysis, booking lead time showed one of the clearest differences between attended and no-show appointments, while several other variables showed weaker individual patterns.

Since your track is analysing HealthConnect from an operational/KPI perspective, I'd like to compare our findings.

Which factors or KPIs in your analysis appear most strongly associated with appointment attendance or no-shows?

I'm particularly interested in whether your findings around booking lead time, reminders, appointment day/time, previous no-shows, distance, or waiting time support what I observed in the modelling work.

Your findings could help me validate the feature choices and interpretation of my baseline model.

Thanks in advance



Fehintoluwa Yesterday at 9:43 PM

Hi Alia, I'm a data analyst intern

From my analysis, the KPIs that stood out were reminder status and previous no-show history, as both showed an association with no-show rates. Booking lead time also showed a relationship with attendance rates.



Alia Al-Qadri Yesterday at 10:59 PM

@Fehintoluwa Thank you 😊

Figure 16. Cross-Track Collaboration with Data Analytics Intern

17. Assumptions

The baseline modelling approach assumes that the historical appointment information is reasonably accurate.

It also assumes that the selected predictors would be available before the appointment outcome occurs.

The model assumes that historical patterns may provide useful information about future appointment attendance.

The current model is intended for administrative decision support only.

It is not intended to make medical or clinical decisions.

18. Limitations

Several limitations remain.

The dataset is fictional and anonymized.

The results therefore cannot automatically be generalized to real healthcare populations.

Some variables required missing-value treatment.

Several predictors showed only weak individual relationships with appointment outcome.

The Logistic Regression model also assumes relatively simple relationships between predictors and the probability of a no-show.

The baseline achieved only moderate predictive performance.

Approximately 35% of actual no-shows were still missed.

The model therefore requires further development before any practical use could be considered.

19. Risks and Dependencies

Data leakage remains an important risk in future development.

Patient-level splitting should continue to be used when evaluating models because patients can have repeated appointments.

Preprocessing and feature transformations should also be learned from training data rather than the complete dataset.

Potential bias should be assessed before demographic variables are used operationally.

The final usefulness of the model will also depend on how its predictions are used by the clinic.

A no-show prediction should support patient engagement and administrative planning.

It should not be used to restrict access to care.

20. Recommendations for Week 6

Week 6 should focus on improving and validating the baseline solution.

Recommended next steps include:

- Compare Logistic Regression with additional classification algorithms.
- Explore models such as Random Forest.
- Perform further feature selection.
- Investigate additional meaningful feature engineering.
- Evaluate model performance using consistent patient-level validation.
- Examine feature importance and model interpretation.
- Consider decision-threshold adjustment based on the importance of detecting no-shows.
- Assess potential fairness concerns.
- Compare improved models against the Week 5 baseline.

The Week 5 Logistic Regression results should remain the reference benchmark.

Any future model should demonstrate a meaningful improvement before replacing the baseline.

21. Conclusion

Week 5 moved the HealthConnect Data Science project from problem definition into practical machine learning development.

The appointment dataset was prepared, explored, and transformed for modelling.

Meaningful features were engineered and assessed for redundancy.

A patient-level train-test strategy was used to reduce the risk of leakage between training and testing data.

A Logistic Regression baseline model was then developed.

The model achieved 63% accuracy, 65% recall for no-shows, a No-Show F1-score of 0.64, and a ROC-AUC of 0.68.

These results demonstrate that the available appointment data contains useful predictive information.

However, the performance remains moderate and should be improved before considering practical application.

Week 5 therefore provides a clear and reproducible baseline for the next stage of the HealthConnect project.